Assignment: 4
Agile devops and development lab

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Create an HTML file for your profile and complete the following:

1. Create a repository in GIT and create the following files:

- **index.html**
- **Dockerfile** • **Deployment.yaml**

index.html:

```
<!DOCTYPE html>
<html>
<head>
<title>CI/CD Demo</title>
<style>
body{
text-align:center; font-
family:Times New Roman;
background-color:#af0d0d;
}
h1{ color:#0ef
777; }
</style>
</head>
<body>
<h1>CI/CD Pipeline Working</h1>
<h2>Deployed using Jenkins + Docker + Kubernetes</h2>
<p>This webpage is automatically deployed using a CI/CD pipeline.</p>
</body>

</html>
```

dockerfile:

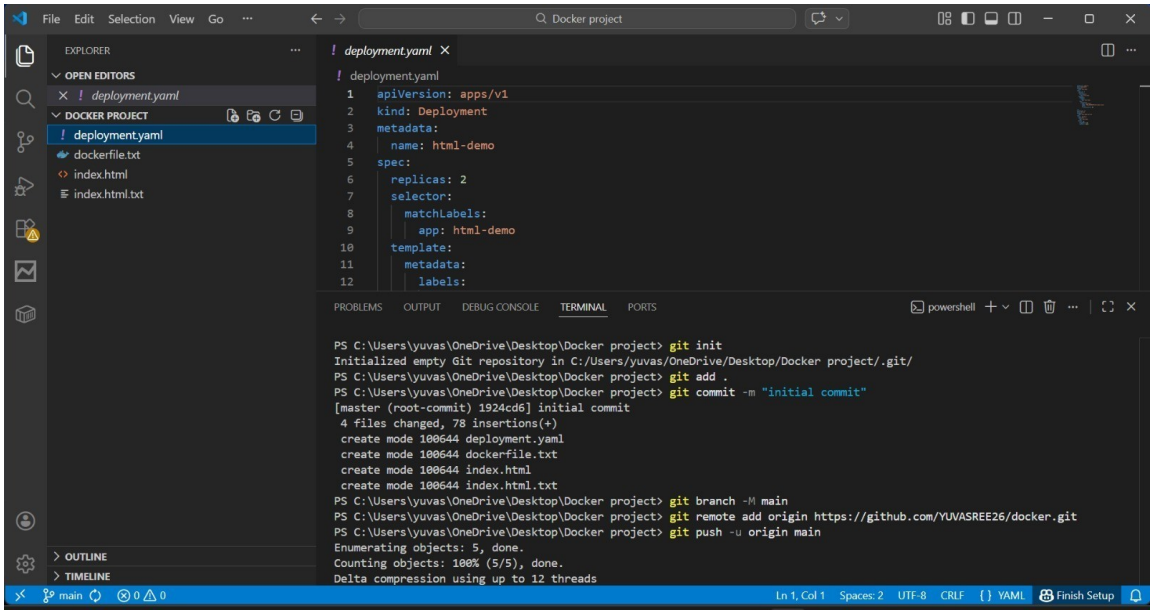
```
FROM nginx:latest
COPY index.html /usr/share/nginx/html/index.html

EXPOSE 80
```

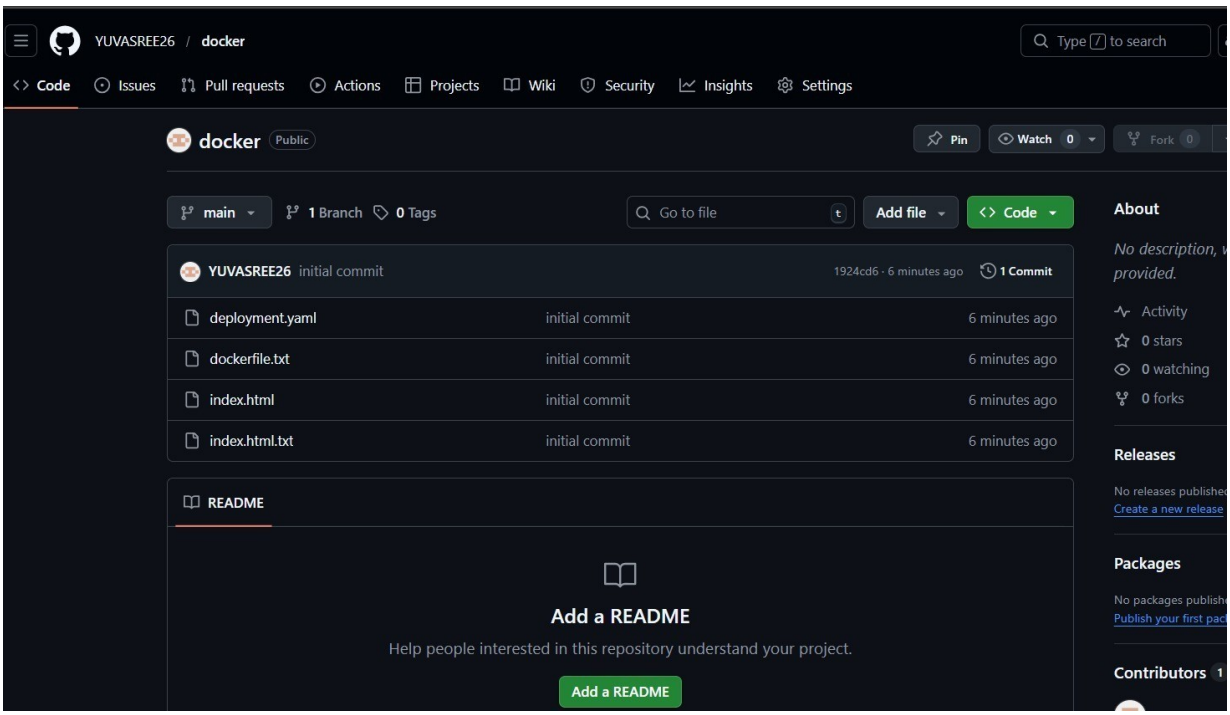
deployment.yaml:

```
apiVersion: apps/v1
kind: Deployment
```

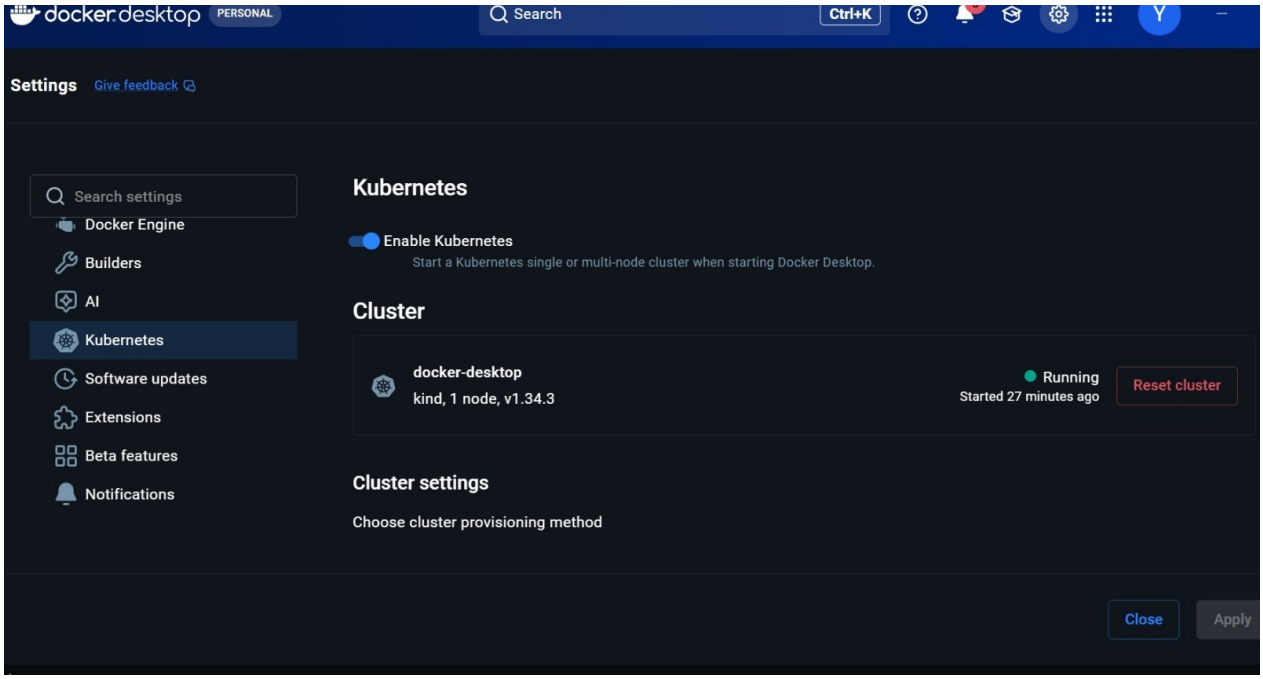
metadata: name:
html-demo spec:
  replicas: 2
selector:
matchLabels:
app: html-demo
template:
metadata: labels:
app: html-demo
spec: containers:
- name: html-demo image: yuva26/html-demo:latest
  ports:
- containerPort: 80

--apiVersion: v1 kind:
Service metadata: name:
html-demo-service spec:
type: NodePort selector:
  app: html-demo
ports: - port: 80
targetPort: 80
nodePort: 30010



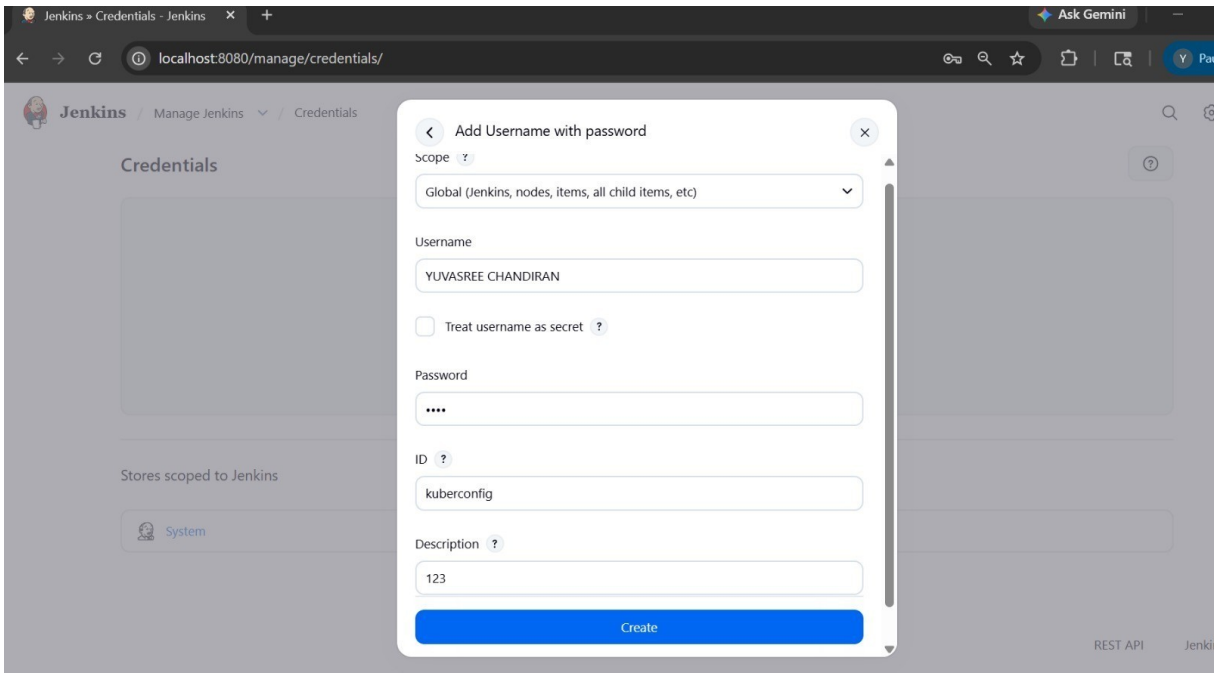


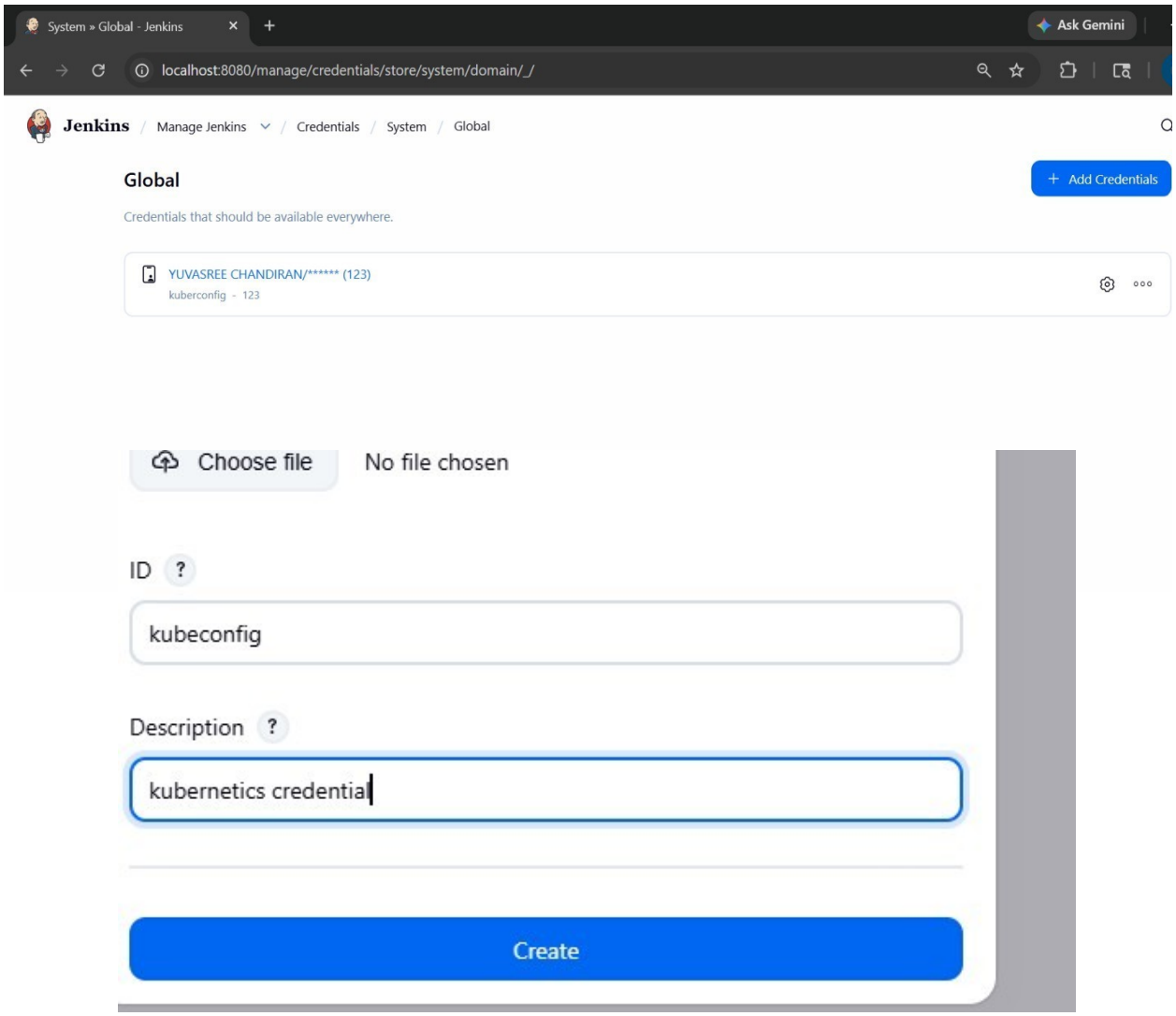
2.Enable Kubernetes



3.Open Jenkins

In Jenkins Setting --> Credentials

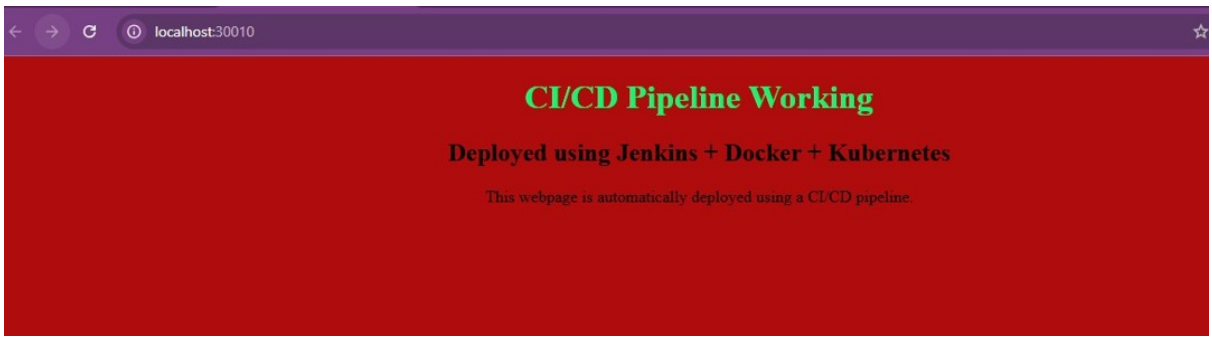




pipeline
{ agent any
environment {
DOCKER_IMAGE = "yuva26/html-demo"

```
}
stages {  stage('Clone
Code') { steps { git
branch: 'main',
url: 'https://github.com/YUVASREE26/docker.git'
}
}
stage('Build Docker Image')
{ steps {
bat 'docker build -t %DOCKER_IMAGE% .'
} } stage('Push Image') { steps
{ withCredentials([usernamePassword(credentialsId:
'dockerhub', usernameVariable: 'USER', passwordVariable:
'PASS')]) { bat 'docker login -u %USER% -p %PASS%' bat
'docker push %DOCKER_IMAGE%'
}
}
}
stage('Deploy to Kubernetes') { steps
{ withCredentials([file(credentialsId: 'kuberconfig',
variable:
'KUBECONFIG')]) {
bat '''
set KUBECONFIG=%KUBECONFIG%
kubectl apply -f deployment.yaml --validate=false
'''
}
}
}
}
}
```

cicd demo

Stage View

	Clone Code	Build Docker Image	Push Image	Deploy to Kubernetes
Average stage times: (full run time: ~22s)	1s	3s	4s	169ms
#3 Mar 26 15:26 1 commit	1s	8s	12s	392ms